

CHAPTER 3

COMBUSTION FUNDAMENTALS AND COMBUSTION CHAMBER PERFORMANCE

3.1. Combustion Fundamentals

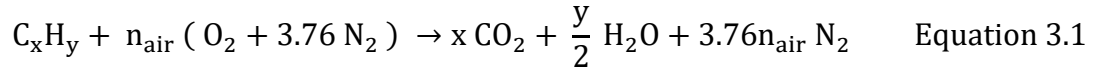
Combustion can be described, in a simplified way, as an exothermic reaction of a fuel and an oxidant. However, this process is extremely complex as it depends on many factors. For example, this process can occur in many forms (or regimes), not all of which accompanied by flame or luminescence. Thus, due to the difficulty in cover all that information and for a better understanding of the reader, in this section, the focus will be made in the combustion processes and parameters that are typically associated to the gas turbine engines.

3.2. Combustion Flames

In gas turbine combustors, flame processes occur in the deflagration regime, where a flame propagates through the unburned mixture. A flame is a rapid chemical reaction in a thin fluid layer, with steep temperature and species concentration gradients, often accompanied by luminescence. In gas turbines, flames are stabilized at a fixed point with a continuous flow of the fuel-air mixture. Flames can be premixed (fuel and air mixed before combustion) or diffusion (mixed in the flame zone). Both types can be laminar or turbulent, depending on flow velocities. In turbines, air is the oxidant, and fuel can be gaseous or liquid. If both reactants are in the same state, the combustion is homogeneous. With liquid fuels, incomplete vaporization can lead to heterogeneous spray combustion, increasing pollutant emissions. This study focuses on diffusion flames, where the mixing rate of fuel and oxidant often limits combustion speed. In diffusion flames, fuel and oxidizer mixing typically occurs through molecular diffusion, making the process slower than in premixed flames. As a result, diffusion flames often produce higher amounts of soot and nitrogen oxides (NO_x), which are significant contributors to environmental pollution. Advances in fuel injection systems aim to improve mixing efficiency, reducing emissions and enhancing overall combustion performance.

3.3. Combustion Stoichiometry

A stoichiometric reaction is one in which the oxidizer is exactly the amount required to completely burn the fuel in the system. It is theoretically the most efficient reaction. Often, for hydrocarbon fuel, the stoichiometric reaction is written in terms of 1 mole of fuel reacting with air, as shown in Equation (3.1).



Where, $n_{air} = x + y/4$ and the constant 3.762 comes from the ratio of approximately 79% nitrogen to 21% oxygen by volume (or 77% nitrogen and 23% oxygen by mass) in the air. This means that for every mole of oxygen required for combustion, 3.762 mol of N_2 must also be introduced. The N_2 does not alter significantly the O_2 balance, however it has a major impact on thermodynamics, chemical kinetics, and formation of pollutants in combustion systems. In this case, the small amounts of CO_2 , argon and other traces of species present in the composition of the oxidizer (assuming it's air) can be disregarded.

For each fuel, the combustion reaction can be classified within two types, complete and incomplete reaction. The difference between these two types of combustion lies in the products of the reaction. In the complete combustion (of a hydrocarbon fuel) are released only completely oxidized products (such as CO_2 and H_2O), and energy while in the incomplete reaction are released completely oxidized products (CO_2 and H_2O), intermediate products (CO, unburned hydrocarbons, etc.), and energy. Based on Equation (3.1), the stoichiometric air to fuel ratio can be determined through the Equation (3.2).

$$(AFR)_{stoich} = \left(\frac{m_{air}}{m_{fuel}} \right)_{stoich} = \frac{4.762 n_{air}}{1} \frac{m_{air}}{m_{fuel}} \quad \text{Equation 3.2}$$

Where, $(AFR)_{stoich}$ = stoichiometric air fuel ratio of combustion

m_{air} = molecular weight of air (g/mol)

m_{fuel} = molecular weight of jet fuel (g/mol)

In practice, however, there is no assurance that simply by providing the correct air to fuel ratio the combustion will be completed.

3.4. Enthalpy of Formation

The enthalpy of formation of a substance, represented as ΔH_f , defined as the heat evolved when 1 mole of the substance is formed from its elements in the respective standard state. The standard state temperature and pressure are considered as $T_{ref}=25^\circ\text{C}$ and $P_{ref}=1\text{ atm}$, respectively, which are consistent with CHEMKIN and NASA thermodynamic databases. For the elements that are in their natural state, the enthalpy of formation is conventionally adopted as zero and the values for the enthalpy of formation for other state than the reference are usually tabulated in the literature.

3.5. Enthalpy of Combustion

The enthalpy of combustion consists in the heat released when a fuel (usually a hydrocarbon) reacts with an oxidant to yield the combustion products. This amount of heat can be calculated relating the absolute enthalpies of the reactants and products as given by Equation (3.3).

$$\Delta H_{\text{combustion}} = \Delta H_{\text{products}} - \Delta H_{\text{reactants}} \quad \text{Equation 3.3}$$

Where, $\Delta H_{\text{combustion}}$ = total enthalpy of combustion

$\Delta H_{\text{products}}$ = total enthalpy products

$\Delta H_{\text{reactants}}$ = total enthalpy of reactants

3.6. Heat Release Rate

The heat release rate (Q) is a measure of the thermal energy produced per unit time during combustion. Expressed in kilowatts (kW), Q is calculated by multiplying the fuel's mass flow rate by its enthalpy change of combustion. This parameter is crucial for evaluating the efficiency and performance of combustion systems in various applications, including engines and industrial furnaces.

$$Q = \dot{m} \times \Delta H_{\text{comb}} \quad \text{Equation 3.4}$$

Where, Q = heat release rate in kJ/s (kw)

$\dot{m}$ = mass flow rate of jet fuel in kg/s

ΔH_{comb} = enthalpy change of combustion per mole of jet fuel kJ/kg

3.7. Combustion Chamber Performance

A combustion chamber must be able to allow fuel to burn efficiently over a wide range of operating conditions without incurring neither in a large pressure loss, nor in any kind of penalty for the structural integrity of the engine components. So, to evaluate the combustion chamber performance, the following parameters/conditions can be used.

- i. Pressure Loss
- ii. Combustion Intensity
- iii. Combustion Efficiency

3.8. Pressure Loss

Pressure loss occurs in any flow process and impacts work output and efficiency in a Joule-Brayton cycle. To minimize this in the combustor, losses are categorized as cold or hot. Cold losses come from expansion, wall friction, and mixing, measured with air flow only. Hot losses, due to temperature rise, are usually smaller. Saravanamuttoo et al. stated that pressure loss, expressed in terms of dynamic head, remains consistent across Reynolds numbers. The Pressure Loss Factor (PLF) is calculated using Equation (3.5), accounting for cold and hot losses separately.

$$\text{Pressure loss factor (PLF)} = \frac{\Delta p_0}{m^2 / 2\rho_1 A_m^2} = K_1 + K_2 \left(\frac{T_{01}}{T_{02}} - 1 \right) \quad \text{Equation 3.5}$$

where, Δp_0 = pressure loss in combustion chamber

m^2 = air mass flow rate

ρ_1 = inlet density

K_1 and K_2 are determined by measurement of the performance of the combustion chamber on a test rig from a cold run and a hot run.

Table 3.1. Typical pressure loss factor in combustion chamber

TYPES OF CHAMBERS	PLF
ANNULAR	20
TUBO-ANNULAR	28
TUBULAR	37

3.9. Density of Air

The density of air in a combustion chamber can be defined as the mass per unit volume of the air present in the chamber. It is a critical parameter in combustion calculations, as it influences the combustion process, pressure, temperature, and the overall performance of the engine.

$$\rho = \frac{P}{R \times T} \quad \text{Equation 3.6}$$

Where, P = pressure of air entering in the combustion chamber

R = the specific gas constant for air , approximately 287(J/kg. K)

T =temperature of the air entering in the combustion chamber

3.10. Combustion Intensity

Combustion intensity is a fundamental parameter in designing combustion chambers for aero gas turbine engines. It quantifies the maximum rate of reactant conversion into heat within a given chamber volume. High combustion intensity signifies that the chamber can release heat at a rapid pace, which is crucial for achieving substantial power output. This becomes particularly significant in aerospace engineering, where there is a constant push to reduce engine size while maintaining high power. By ensuring a high combustion intensity, engineers can design more compact chambers that still provide the necessary power. This capability is essential for optimizing the power-to-weight ratio and enhancing overall engine efficiency within limited spatial constraints.

$$\begin{aligned} \text{Combustion Intensity (I}_c\text{)} &= \frac{Q}{V \times P_3} \\ &= \frac{Q}{A_{cc} \times L_{cc} \times P_3} \end{aligned} \quad \text{Equation 3.7}$$

Where, Q = heat release rate

V = volume of the combustion chamber

A_{cc} = total area of the combustion chamber

L_{cc} = total length of the combustion chamber

P₃ = combustor exit pressure

3.11. Total Area of Annular Combustor

The total area of an annular combustor refers to the surface area through which the combustion gases flow within the combustor. This area is crucial for determining parameters like airflow, fuel distribution, heat release, and pressure loss in the combustion process.

In an annular combustor, the flow area is typically the difference between the outer and inner cylindrical surfaces of the annular space. This can be visualized as the area between two concentric cylinders.

$$A_{\text{total}} = \int_0^L \pi \times \left(\frac{D_{\text{out}}}{2} - \frac{D_{\text{in}}}{2} \right) dr \quad \text{Equation 3.8}$$

where, A_{total} = total area of the combustion chamber

D_{out} = outer diameter of the combustion chamber

D_{in} = inner diameter of the combustion chamber

L = length of the annular combustor

3.12. Combustion Efficiency

Combustion efficiency is a key performance indicator in the evaluation of a combustor's effectiveness. It quantifies the combustor's ability to convert the chemical energy stored in the fuel into thermal energy, which is then transferred as heat to the working fluid, comprising air and combustion products. This efficiency is of paramount importance, as it directly impacts the overall performance of the engine, influencing both fuel economy and emissions.

The concept of combustion efficiency is expressed as the ratio of the actual heat release to the ideal heat release during the combustion process. The actual heat release refers to the amount of thermal energy that is effectively transferred to the working fluid, while the ideal heat release represents the maximum possible thermal energy that could be obtained under perfect combustion conditions. A high combustion efficiency indicates that the combustor is operating close to its ideal conditions, effectively utilizing the fuel's energy potential.

Achieving high combustion efficiency is crucial in engine design, particularly in the context of aero engines, where maximizing energy conversion and minimizing fuel consumption are essential. By ensuring that the majority of the fuel's chemical

energy is converted into useful thermal energy, the engine can operate more efficiently, leading to improved performance, reduced operational costs, and lower environmental impact. Mathematically, combustion efficiency can be expressed as:

$$\begin{aligned}\eta_b &= \frac{(\Delta T_0)_{\text{actual}}}{(\Delta T_0)_{\text{ideal}}} \\ &= \frac{(T_{02} - T_{01})_{\text{actual}}}{(T_{02} - T_{01})_{\text{ideal}}}\end{aligned}\quad \text{Equation 3.9}$$

Where, η_b = Combustion efficiency

$(\Delta T_0)_{\text{actual}}$ = Actual temperature rise in the combustion chamber

$(\Delta T_0)_{\text{ideal}}$ = Ideal temperature rise in the combustion chamber